



Learning Objectives

- ▶ By the end of this lecture, students will be able to:
- ▶ Define Decision Support Systems and explain their purpose.
- ▶ Describe the components and architecture of a DSS.
- ▶ Identify types of DSS and their applications.
- ▶ Explain how DSS supports managerial decision making.
- ▶ Discuss advantages and limitations of DSS.

What Is a Decision?

- ▶ A **decision** is a choice made among alternatives to achieve an objective.
- ▶ Types of decisions:
- ▶ **Structured** - routine, repetitive, programmed
- ▶ **Semi-structured** - partly routine, partly judgment-based
- ▶ **Unstructured** - complex, non-routine, requiring human judgment

What Is a DSS?

- ▶ **Definition:**

A **Decision Support System** is a computer-based information system that helps managers make **semi-structured and unstructured decisions** by providing data, models, and analysis tools.

- ▶ **Purpose:**

- ▶ Improve decision quality
- ▶ Increase speed of decision-making
- ▶ Support, NOT replace, human judgment

Characteristics of DSS

- ▶ Interactive and user-friendly
- ▶ Flexible and adaptable
- ▶ Supports semi-structured/unstructured decisions
- ▶ Uses analytical models
- ▶ Uses data from internal and external sources
- ▶ Supports “what-if” and simulation analysis

Components of a DSS

Database (Data Management Subsystem)

- ▶ Stores internal and external data
- ▶ Includes historical data, real-time data, and metadata

Model Base (Model Management Subsystem)

- ▶ Includes mathematical, statistical, and simulation models
- ▶ Examples: forecasting models, optimization models

User Interface (UI)

- ▶ Tools that allow managers to interact with the system
- ▶ Dashboards, forms, reports

Knowledge Engine / Intelligent Component (optional)

- ▶ Uses AI to enhance decision making
- ▶ Expert systems, rule-based engines

DSS Architecture

- ▶ A typical DSS includes:
- ▶ Data sources (internal & external)
- ▶ DSS database
- ▶ Model base
- ▶ User interface
- ▶ Decision maker



Types of DSS

1. Data-Driven DSS

- ▶ Focuses on access to large datasets
- ▶ Example: data warehouses, OLAP systems

2. Model-Driven DSS

- ▶ Uses models to analyze data
- ▶ Example: budgeting, scheduling, simulation tools

3. Knowledge-Driven / Expert DSS

- ▶ Uses AI and rules
- ▶ Example: medical diagnosis support

4. Document-Driven DSS

- ▶ Manages, retrieves, and analyzes documents
- ▶ Example: enterprise search engines

5. Communication-Driven DSS

- ▶ Supports groups working together
- ▶ Example: video conferencing, groupware

Decision Support Techniques

- ▶ What-if analysis
- ▶ Sensitivity analysis
- ▶ Goal-seeking
- ▶ Optimization models
- ▶ Forecasting (time series)
- ▶ Simulation (Monte Carlo)
- ▶ Dashboards and data visualization

Benefits of DSS

- ▶ Better decision quality
- ▶ Faster decision-making
- ▶ Improved productivity
- ▶ Enhanced problem solving
- ▶ Supports teamwork
- ▶ Increased competitiveness
- ▶ Helps with complex and large-scale decisions

Examples of DSS Applications

- ▶ **Banking:** credit scoring, fraud detection
- ▶ **Healthcare:** diagnosis support, treatment planning
- ▶ **Retail:** inventory forecasting, customer analytics
- ▶ **Manufacturing:** production scheduling
- ▶ **Finance:** investment portfolio analysis
- ▶ **Transportation:** route optimization

Questions

- ▶ Why is DSS important for modern organizations?
- ▶ Which type of DSS do you think is most useful today?
- ▶ Can DSS replace managers in decision-making? Why or why not?



Expert Systems

What Is an Expert System?

- ▶ **Definition:**

An **Expert System (ES)** is a computer-based system that uses **knowledge** and **inference rules** to solve complex problems that normally require **human expert reasoning**.

- ▶ **Main Goal:**

To mimic the decision-making ability of a human expert.

Characteristics of Expert Systems

- ▶ Mimic human reasoning
- ▶ Solve complex, domain-specific problems
- ▶ Explain their reasoning (explanation facility)
- ▶ Use symbolic knowledge (rules, facts)
- ▶ High level of accuracy
- ▶ Interactive and user-friendly

When Are Expert Systems Used?

- ▶ Expert systems are useful when:
- ▶ Expert knowledge is needed but not always available
- ▶ Decisions are repetitive and rule-based
- ▶ Problems are complex but follow known patterns
- ▶ Human expertise is expensive or rare
- ▶ Examples: medical diagnosis, equipment repair, troubleshooting.

Components of an Expert System

1. Knowledge Base

- ▶ Contains facts + rules
- ▶ Example rules:
IF fever AND cough THEN flu

2. Inference Engine

- ▶ Applies rules to facts
- ▶ Reasoning methods: Forward chaining, Backward chaining

3. User Interface

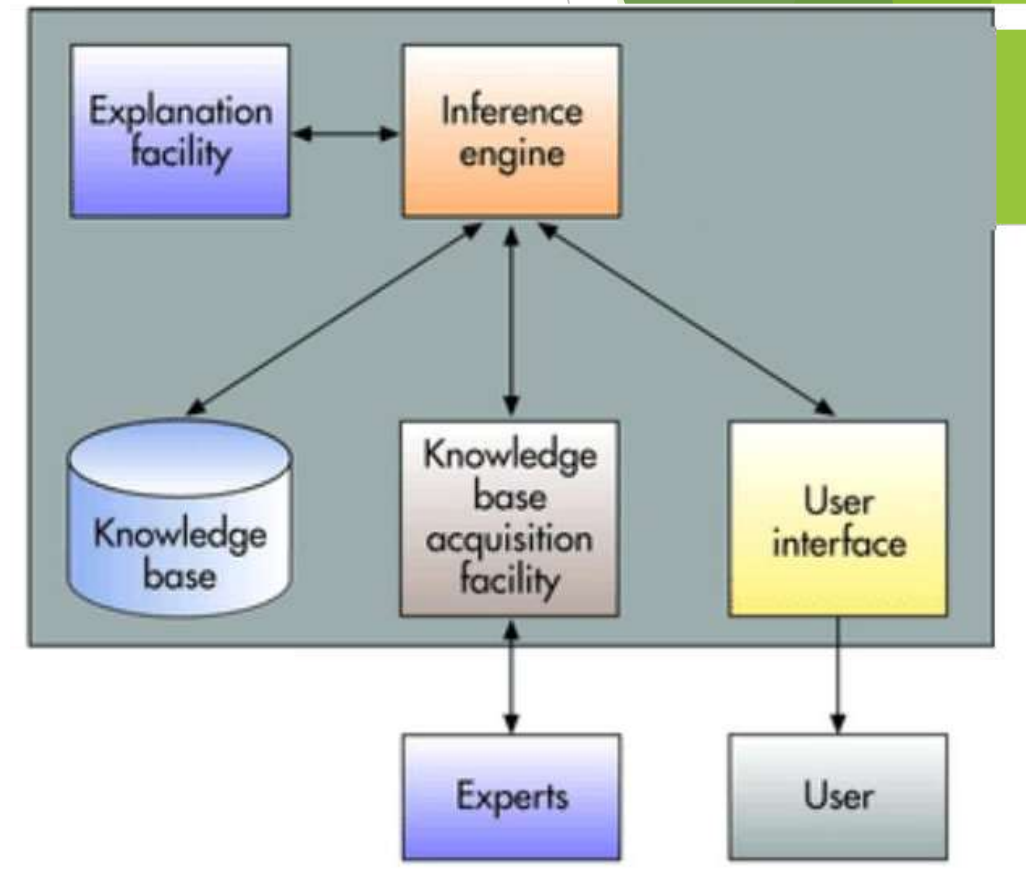
- ▶ Allows users to interact with the system

4. Explanation Facility

- ▶ Explains how conclusions are reached

5. Knowledge Acquisition Module

- ▶ Helps experts add or update knowledge



Knowledge Representation Methods

Expert systems use symbolic forms such as:

- ▶ **Rules (IF-THEN rules)**
- ▶ **Frames (objects with attributes)**
- ▶ **Semantic networks (nodes and links)**
- ▶ **Decision trees**
- ▶ **Logic (propositional or predicate)**

Example Rule:

IF account balance < 0

THEN send overdraft warning

Reasoning Techniques

1. Forward Chaining (Data-Driven)

Starts with available facts → applies rules → reaches conclusion.
Used in: monitoring, diagnostics.

2. Backward Chaining (Goal-Driven)

Starts with a goal → checks rules to prove it.
Used in: medical diagnosis, troubleshooting.

Applications of Expert Systems

► **Medicine**

- Diagnosing diseases (MYCIN, CADIAG)

► **Finance**

- Loan approval
- Credit scoring

► **Manufacturing**

- Fault diagnosis
- Quality control

► **Agriculture**

- Pest identification
- Crop management

► **Customer Support**

- Troubleshooting (printers, routers)

Advantages of Expert Systems

- ▶ Increased accuracy and reliability
- ▶ Consistent decision-making
- ▶ Available anytime (24/7)
- ▶ Preserve rare human expertise
- ▶ Reduce training costs
- ▶ Faster solutions
- ▶ Can handle large knowledge bases



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Data Analytics

Learning Objectives

- ▶ By the end of this lecture, students will be able to:
- ▶ Define Data Analytics and explain its importance.
- ▶ Identify the types of data analytics: descriptive, diagnostic, predictive, and prescriptive.
- ▶ Understand data sources and data quality concepts.
- ▶ Describe tools and technologies used in data analytics.
- ▶ Apply the analytics process and understand real-world business applications.

What Is Data Analytics?

- ▶ **Definition:**
Data Analytics is the process of **inspecting, cleaning, transforming, and modeling data** to extract useful information, discover patterns, and support decision-making.
- ▶ **Objective:**
Turn raw data into **insights** that help organizations make better decisions.

Why Is Data Analytics Important?

- ▶ Helps organizations make evidence-based decisions
- ▶ Improves efficiency and operations
- ▶ Identifies opportunities and risks
- ▶ Enhances customer experience
- ▶ Supports innovation and competitiveness
- ▶ Enables automation and smart systems

Data vs. Information vs. Insights

- ▶ **Data:** Raw facts and figures
- ▶ **Information:** Processed/organized data
- ▶ **Insights:** Actionable understanding derived from data
Example:
Data = sales numbers
Insight = why sales dropped last month

Types of Data Analytics

▶ Descriptive Analytics

- ▶ “What happened?”
Uses past data → reports, dashboards

▶ Diagnostic Analytics

- ▶ “Why did it happen?”
Uses drill-down, root-cause analysis

▶ Predictive Analytics

- ▶ “What will happen?”
Uses statistical models, machine learning

▶ Prescriptive Analytics

- ▶ “What should we do?”
Uses optimization and simulation models

Data Sources

- ▶ Internal business systems .
- ▶ Social media
- ▶ Sensors and IoT devices
- ▶ Websites & clickstream data
- ▶ Mobile apps
- ▶ Government and open data
- ▶ Surveys and customer feedback

The Data Analytics Process

- ▶ Define the problem
- ▶ Collect data
- ▶ Clean and prepare data
- ▶ Analyze data
- ▶ Interpret results
- ▶ Communicate insights
- ▶ Make decisions

Data Preparation (Data Wrangling)

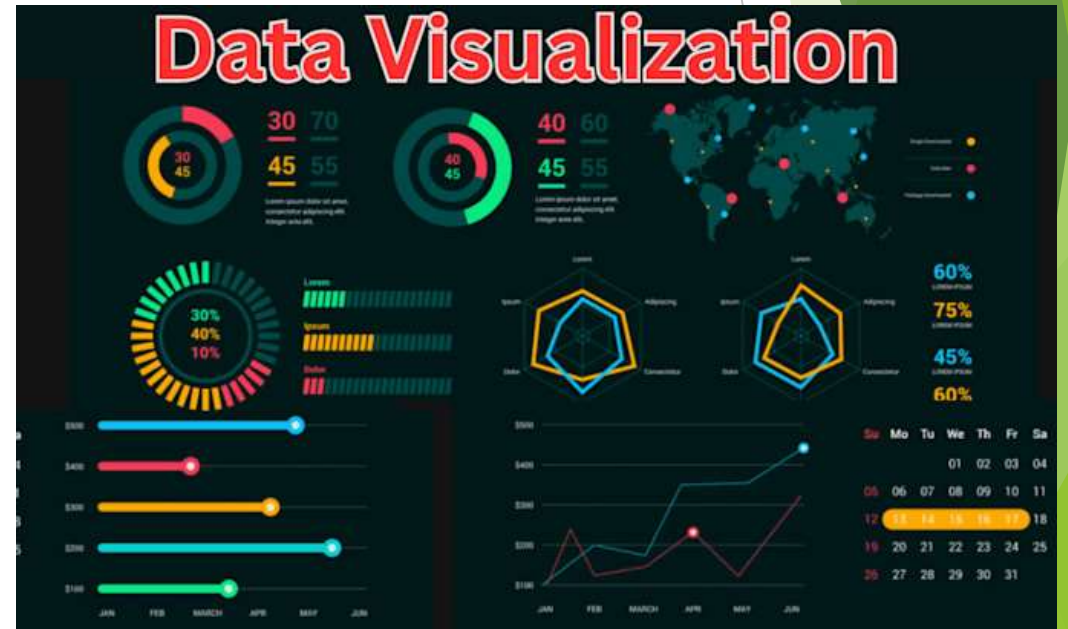
- ▶ Most time is spent here.
- ▶ Tasks:
 - ▶ Removing missing values
 - ▶ Handling duplicates
 - ▶ Formatting and transforming data
 - ▶ Normalization and standardization
 - ▶ Feature engineering (for machine learning)

Analytical Techniques

- ▶ Summary statistics
- ▶ Correlation analysis
- ▶ Trend analysis
- ▶ Regression (linear, logistic)
- ▶ Clustering
- ▶ Classification
- ▶ Time-series forecasting
- ▶ Visualization (charts, heatmaps)

Data Visualization

- ▶ Tools for creating charts to communicate insights:
- ▶ Bar charts
- ▶ Line graphs
- ▶ Pie charts
- ▶ Histograms
- ▶ Scatter plots
- ▶ Dashboards



Tools & Technologies Used in Data Analytics

▶ Basic Tools

- ▶ Excel
- ▶ Google Sheets

▶ Analytics & BI Tools

- ▶ Power BI
- ▶ Tableau

▶ Programming Tools

- ▶ Python (Pandas, NumPy, Matplotlib)
- ▶ R

▶ Big Data Tools

- ▶ Hadoop
- ▶ Spark

▶ Database Systems

- ▶ SQL
- ▶ NoSQL (MongoDB)

Questions

- ▶ Which type of data analytics do you think is most important today?
- ▶ What challenges does an organization face in implementing analytics?
- ▶ What industries benefit most from predictive analytics?